

CLASS 7 MATH

Premium Workbook & Olympiad Prep

Algebra, Rational Numbers, Geometry & 100 Complex Logic Puzzles

Chapter 1: Rational Numbers & Exponents

SECTION A: CORE CONCEPTS

1. Express the rational number $-48/72$ in its standard form.

2. Find three rational numbers between $-1/2$ and $2/3$.

3. Evaluate using laws of exponents: $(3^5 * 10^5 * 25) / (5^7 * 6^5)$.

4. Express 384,000,000 in standard scientific notation.

5. Simplify: $\left[-\frac{2}{3}\right]^3^2$.

SECTION B: REAL WORLD APPLICATIONS

1. The distance from Earth to the Moon is approximately 384,400 km. Express this distance in scientific notation.

2. A computer processor can perform an operation in 1.5×10^{-9} seconds. How many operations can it perform in 3 seconds?

3. Atharv's Jigyasa app database grew by 2^{10} users in the first month and 2^{12} users in the second month. How many total users joined in the first two months?

Chapter 2: Algebraic Expressions & Equations

SECTION A: CORE CONCEPTS

1. Add the expressions: $(5x^2 - 3x + 4)$ and $(2x^2 + x - 7)$.

2. Subtract $(3a - 4b - c)$ from $(5a + 2b - 3c)$.

3. Solve the linear equation for y : $5y/2 - 3 = 12$.

4. Solve for x: $3(x - 4) = 2(2x + 1)$.

5. Evaluate the expression $a^3 - b^3$ when $a = 3$ and $b = -2$.

SECTION B: REAL WORLD APPLICATIONS

1. The length of a rectangular garden is 5 meters more than twice its width. If the perimeter is 70 meters, find its dimensions.

2. Mukesh's age is 3 times Atharv's age. After 10 years, Mukesh will be twice as old as Atharv. Find their present ages.

3. Priyanka bought 'x' packets of Crunchley makhana at \$4 each and 'y' packets at \$5 each. Write an algebraic expression for the total amount spent.

Chapter 3: Comparing Quantities (Percentages & Interest)

SECTION A: CORE CONCEPTS

1. Convert the ratio 3:4 into a percentage.

2. Find 12.5% of 800 kg.

3. An item is bought for \$400 and sold for \$450. Find the profit percentage.

4. Find the simple interest on \$2,500 at 6% per annum for 3 years.

5. If 15 pens cost \$45, what is the cost of 25 pens? (Use unitary method).

SECTION B: REAL WORLD APPLICATIONS

1. Priyanka deposits \$5,000 in a bank that pays 4.5% simple interest per year. What will be her total amount after 4 years?

2. The population of San Jose increased from 950,000 to 1,045,000 over a decade. What was the percentage increase in population?

3. A shop offers a 20% discount on an EGO Power+ mower originally priced at \$550. What is the sale price?

Chapter 4: Geometry & Mensuration

SECTION A: CORE CONCEPTS

1. Find the circumference and area of a circle with a radius of 21 cm. (Use $\pi = \frac{22}{7}$).

2. Find the area of a triangle with a base of 24 cm and a height of 15 cm.

3. A parallelogram has an area of 180 sq cm and a base of 15 cm. Find its corresponding height.

4. In a right-angled triangle, the two shorter sides are 6 cm and 8 cm. Find the length of the hypotenuse.

5. Two lines intersect, and one of the vertically opposite angles is 45° . Find the other three angles.

SECTION B: REAL WORLD APPLICATIONS

1. Mukesh mows a circular lawn with a diameter of 42 meters. What is the total area of the grass he mowed?

2. A picture 40 cm long and 20 cm wide is to be framed with a 2 cm wide border all around it. Find the area of the border.

3. Atharv flies a kite. The string is 50 meters long and the kite is directly above a point 30 meters away from Atharv on the ground. How high is the kite flying?

Multi-Concept Combined Practice

Level 1: Core Proficiency (Average Student) - SET 1

1. Express $-36/48$ as a rational number in its standard form.

2. Find the value of $2^3 \times 3^2$.

3. Solve for x: $3x - 5 = 16$.

4. Find the area of a circle with a radius of 7 cm. (Use $\pi = \frac{22}{7}$).

5. Convert $\frac{3}{4}$ into a percentage.

6. Find 15% of 600.

7. Add the algebraic expressions: $(3a - 4b)$ and $(2a + 5b)$.

8. Find the mean of the first 5 prime numbers.

9. Divide the rational number $-\frac{4}{5}$ by $\frac{8}{15}$.

10. A car travels 240 km in 4 hours. What is its average speed?

Multi-Concept Combined Practice

Level 1: Core Proficiency (Average Student) - SET 2

1. Write 450,000 in standard scientific notation.

2. Evaluate: $(5^2)^3$.

3. Find the circumference of a circle with a diameter of 28 cm.

4. Convert 45% into a fraction in its simplest form.

5. Subtract $(2x - 3y)$ from $(5x + y)$.

6. If the ratio of boys to girls is 4:5 and there are 20 boys, how many girls are there?

7. Find the simple interest on \$1000 at 5% per annum for 2 years.

8. What is the median of the data set: 12, 15, 11, 19, 14?

9. A packet of Crunchley makhana costs \$4.50. How much do 12 packets cost?

10. Solve for m: $m/4 + 7 = 12$.

Multi-Concept Combined Practice

Level 2: Advanced Application (Top 10%) - SET 1

1. Find three rational numbers between $\frac{1}{4}$ and $\frac{1}{2}$.

2. The sum of three consecutive odd integers is 69. Find the integers.

3. If $5^x = 125$, find the value of $2^{(x+1)}$.

4. A jacket originally costs \$120. It is marked down by 25%. What is the new price?

5. Simplify the expression: $4(x - 3) - 2(x + 5)$.

6. The angles of a triangle are in the ratio 2:3:4. Find the measure of all three angles.

7. A person borrows \$2500 at 8% per annum simple interest. How much total amount will he pay back after 3 years?

8. Priyanka drives at 60 km/h for 2 hours, then at 90 km/h for 2 hours. What is her average speed for the whole trip?

9. If $x = 2$ and $y = -3$, find the value of $3x^2 - 2xy + y^2$.

10. Find the area of a parallelogram with a base of 18 cm and a corresponding height of 12 cm.

Multi-Concept Combined Practice

Level 2: Advanced Application (Top 10%) - SET 2

1. Atharv bought a book for \$15 at a 20% discount. What was the original price of the book?

2. Express $(2^3 \times 3^4 \times 4) / (3 \times 32)$ in exponential form.

3. A recipe calls for $\frac{3}{4}$ cup of sugar to make 15 cookies. How much sugar is needed for 40 cookies?

4. Find the value of x if: $2(x + 3) = 3(x - 2)$.

5. What is the cost of fencing a circular park of radius 35m at the rate of \$10 per meter?

6. A shopkeeper sells an item for \$450, making a profit of 12.5%. What was the cost price?

7. Two supplementary angles are such that one is 30 degrees more than twice the other. Find the angles.

8. If a train 250 meters long crosses a pole in 12 seconds, what is its speed in km/h?

9. Find the mode and range of: 5, 8, 5, 12, 14, 8, 8, 10.

10. The area of a square is 196 sq cm. Find its perimeter.

Multi-Concept Combined Practice

Level 3: Logical Reasoning (Top 1%) - SET 1

1. If $(x/2) - (1/4) = (x/3) + (1/2)$, find the value of x .

2. A water tank is $3/8$ full. After adding 25 liters, it is $4/5$ full. What is the total capacity of the tank?

3. Find the smallest number which when divided by 12, 15, and 18 leaves a remainder of 5 in each case.

4. A digital clock shows 04:20. What is the exact angle between the hour and minute hands? (Remember the hour hand moves).

5. If 8 workers can build a wall in 15 days, how many days will 5 workers take to build the same wall?

6. The price of a laptop is increased by 25%, then decreased by 20%. What is the net percentage change in its price?

7. Find the sum of all prime numbers between 50 and 70.

8. A wheel has a radius of 35 cm. How many revolutions will it make to travel 1.1 km? (Use $\pi = \frac{22}{7}$).

9. Evaluate: $(2^{-1} + 3^{-1})^{-1}$.

10. I am a 2-digit number. The sum of my digits is 11. If you add 27 to me, my digits are reversed. Who am I?

Multi-Concept Combined Practice

Level 3: Logical Reasoning (Top 1%) - SET 2

1. Mukesh invests \$1200. Part of it is at 5% simple interest and the rest at 7% simple interest. If the total annual interest is \$76, how much was invested at each rate?

2. In a school, the ratio of boys to girls is 5:3. If 20 more boys and 20 more girls join, the ratio becomes 3:2. How many students were originally in the school?

3. Simplify: $\frac{1}{(1 \cdot 2)} + \frac{1}{(2 \cdot 3)} + \frac{1}{(3 \cdot 4)} + \dots + \frac{1}{(9 \cdot 10)}$.

4. A solid cube of side 8 cm is painted red on all faces, then cut into 2 cm cubes. How many small cubes have exactly one red face?

5. Evaluate: $[(1/3)^{-2} - (1/2)^{-3}] \div (1/4)^{-2}$.

6. The average age of 6 students is 14 years. If the teacher's age is included, the average becomes 18. How old is the teacher?

7. Pipe A can fill a tank in 10 hours, and Pipe B can empty it in 15 hours. If both are opened simultaneously, how long will it take to fill the empty tank?

8. Find the number of diagonals in a regular dodecagon (12-sided polygon).

9. If $x : y = 3 : 4$ and $y : z = 8 : 9$, find the ratio $x : y : z$.

10. A rectangular sheet of paper 88 cm long and 20 cm wide is rolled along its length to form a cylinder. What is the volume of the cylinder? (Use $\pi = \frac{22}{7}$).

Multi-Concept Combined Practice

Level 4: Critical Thinking (Top 0.1%) - SET 1

1. Solve the cryptarithm: $TWO + TWO = FOUR$. Find the values of T, W, O, F, U, R. (Assume $O=4$, find the rest).

2. Find the last digit (units digit) of 7^{105} .

3. You have a 5-liter jug and a 7-liter jug. How do you measure exactly 6 liters of water? Explain the steps.

4. A snail is at the bottom of a 50-foot well. Every day it climbs 5 feet, but slips down 3 feet at night. On which day does it escape the well?

5. How many zeros are there at the end of 200 factorial (200!)?

6. Two trains, 300 km apart, are traveling towards each other at 60 km/h and 40 km/h. A bird flies back and forth between them at 80 km/h until they crash. How far did the bird fly?

7. Find the Greatest Common Divisor (GCD) of $2^{100} - 1$ and $2^{120} - 1$.

8. How many triangles are there in a regular hexagon with all its diagonals drawn?

9. A man has 60 socks: 20 blue, 20 black, 20 red. It is completely dark. How many must he pull out to ensure he has a matching pair?

10. Use four 5s and any mathematical operators (+, -, *, /, !, ^) to make the number 24.

Multi-Concept Combined Practice

Level 4: Critical Thinking (Top 0.1%) - SET 2

1. You have 12 identical-looking coins. One is counterfeit and has a different weight (you don't know if it's heavier or lighter). Using a balance scale exactly 3 times, how do you find it?

2. Find the smallest positive integer that leaves a remainder of 2 when divided by 3, 3 when divided by 4, 4 when divided by 5, and 5 when divided by 6.

3. Three travelers pay \$30 for a room. The manager refunds \$5 to the bellhop. The bellhop keeps \$2 and gives \$1 to each traveler. They paid \$9 each (\$27) + \$2 kept by bellhop = \$29. Where is the missing dollar?

4. If 1.5 chickens lay 1.5 eggs in 1.5 days, how many eggs will 9 chickens lay in 9 days?

5. A hollow sphere has an external volume of 1000 cm^3 and a thickness of 2 cm. What is the volume of the cavity inside? (Express in terms of π).

6. Find the sum of the interior angles of a 20-sided polygon.

7. I am a 3-digit number. My hundreds digit is thrice my tens digit. My ones digit is the difference between my hundreds and tens digit. What are the possible numbers?

8. Evaluate the infinite nested square root: $\sqrt{6 + \sqrt{6 + \sqrt{6 + \dots}}}$

9. A grandfather clock takes 12 seconds to strike 5 o'clock. How many seconds will it take to strike 10 o'clock?

10. What is the next number in the sequence: 1, 2, 6, 21, 88, 445, ... ?

Multi-Concept Combined Practice

Level 5: National Olympiad (Top 10) - SET 1

1. Einstein's Riddle: 5 houses, 5 colors, 5 nationalities, 5 drinks, 5 pets. If the Brit lives in the red house, the Swede has dogs, the Dane drinks tea, the green house is just left of the white house... (State the logical steps to determine who owns the fish).

2. Find the number of trailing zeros in the product of the first 100 prime numbers.

3. 100 prisoners are in solitary confinement. A warden will randomly bring them one at a time to a room with a light switch. They can toggle the switch or leave it. They can assert 'everyone has visited the room'. If right, they go free. If wrong, they die. What is the winning strategy?

4. A magical lily pad doubles in size every day. It covers a pond in 40 days. On what day did it cover $\frac{1}{8}$ th of the pond?

5. How many 4-digit numbers contain the digit '7' exactly once?

6. Four people cross a bridge at night with one flashlight. They take 1, 2, 5, and 10 minutes. Max 2 people can cross at a time. What is the shortest time for all to cross?

7. A king has 1000 wine bottles, 1 is poisoned (takes 24 hours to act). He has 10 mice. How can he find the poisoned bottle in 24 hours?

8. What is the maximum number of regions a circle can be divided into by drawing 5 straight lines?

9. If $a^b = b^a$ and $a \neq b$, find a positive integer solution for (a, b) .

10. How many total squares of any size can you count in an 8x8 chessboard?

Multi-Concept Combined Practice

Level 5: National Olympiad (Top 10) - SET 2

1. The sum of the digits of a 3-digit number is 15. If the digits are reversed, the number is decreased by 396. The tens digit is the average of the hundreds and ones digit. What is the number?

2. A monkey climbs a 100 ft pole. He goes up 4 ft in one minute, and slips down 3 ft the next minute. How many minutes to reach the top?

3. Car A and Car B race. Car A travels half the total distance at 40mph and half at 80mph. Car B travels half the total time at 40mph and half the time at 80mph. Who wins the race?

4. You have a 4-minute hourglass and a 7-minute hourglass. How do you measure exactly 9 minutes?

5. What day of the week will it be exactly 10^{10} days from today, if today is Thursday?

6. There are 25 horses. You can race 5 at a time, no stopwatch. What is the absolute minimum number of races needed to find the top 3 fastest horses?

7. In a tournament of 1024 players (single elimination), how many total matches are played to find the winner?

8. A cube is made of 1000 smaller $1 \times 1 \times 1$ cubes. You drill a hole straight through the center from the front face to the back face (a 1×1 column). How many $1 \times 1 \times 1$ cubes are left?

9. If $A=1$, $B=2$... $Z=26$, what is the mathematical product of the letters in the word 'MATH'?

10. Find a 10-digit number where the first digit tells how many 0s are in the number, the second tells how many 1s... up to 9s.
